

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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NTA211-R

Top 10 best practices for AWS beginners

Keith Wood

(he/him)

Principal Solutions Architect

AWS

Renato Fichmann

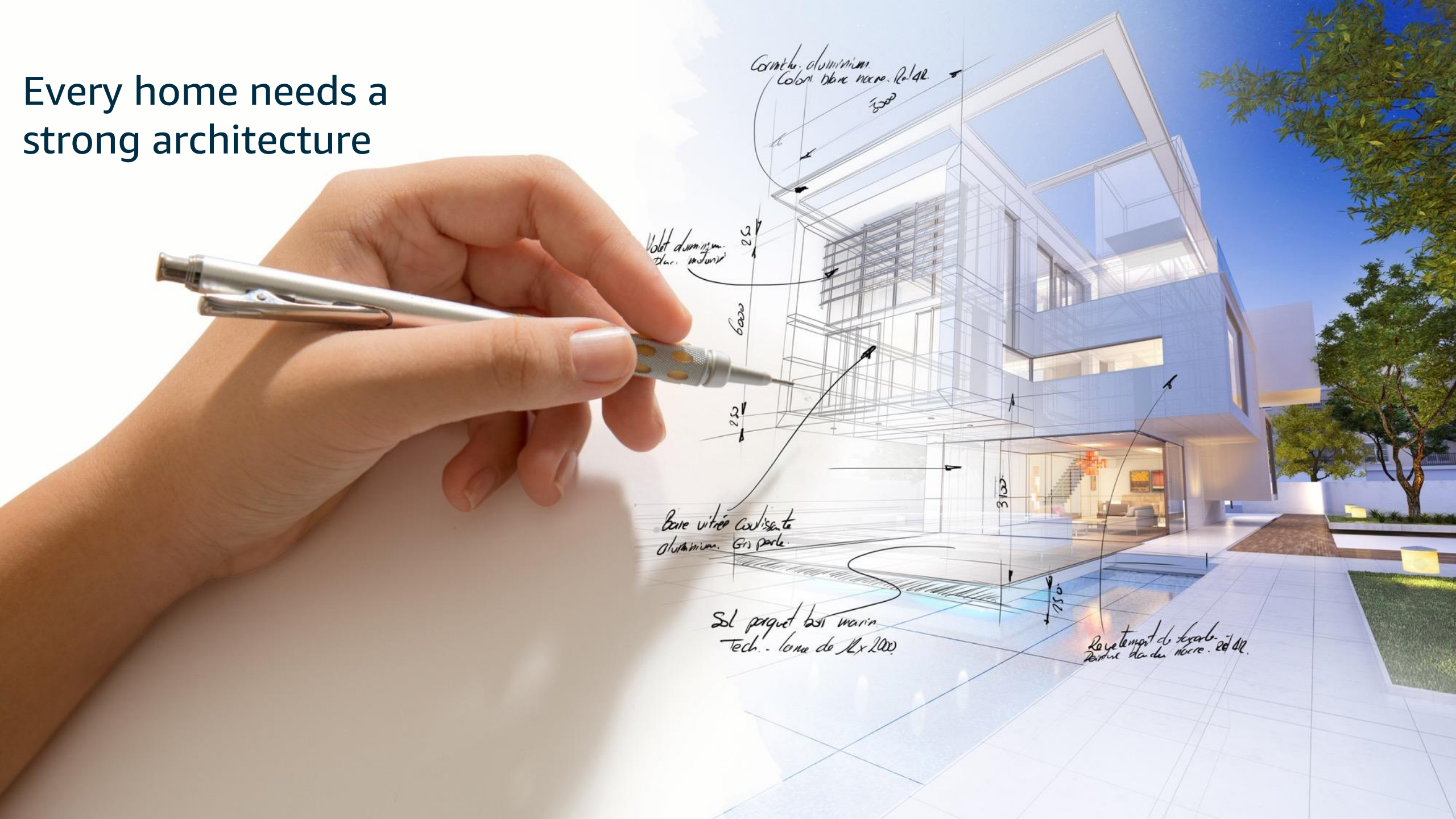
(he/him)

Senior Solutions Architect

AWS



Every home needs a strong architecture



Comble aluminium
Color blanc noir. R/42
3000

Volet aluminium
dure. motorisé

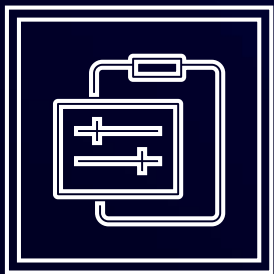
25
base
25

Base vitrée coulissante
aluminium. Gris perle.

Sol parquet bois marin
Tech. - lona de 12x2000

Ravetement de l'escalier
Reinforce de la couleur. 20/42

Top interests for customers new to AWS



Management and
governance

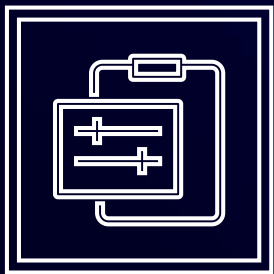


Security and
networking



Application
modernization

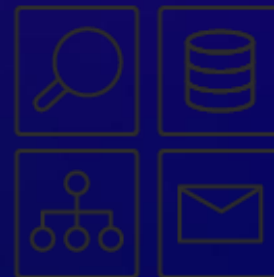
Top interests for customers new to AWS



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Security and
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Application
modernization

Create a landing zone

Related AWS service



AWS Control Tower

Implements services such as **IAM Identity Center** and **AWS Organizations** that help you manage a multi-account environment

Automates AWS Account creation

Provides **AWS best practice blueprints and guardrails**

Effectively manage EC2 instances

Related AWS service



AWS Systems Manager

Activate Systems Manager for your **entire Organization**

Use the **Patch Manager** feature to roll out OS-level patches on an ongoing basis

Use the **Fleet Manager** feature to perform common administrative tasks, including browser-based remote desktop and SSH

Closely monitor costs

Related AWS service



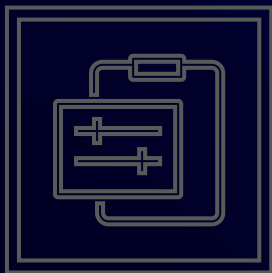
AWS Budgets

Set **custom budgets** for cost and usage thresholds

Receive regular **budget reports**, and **be alerted** when cost and usage exceed (or are forecasted to exceed) your budgeted amount

Execute actions automatically or via a workflow when a budget target has been exceeded or is forecasted to be exceeded

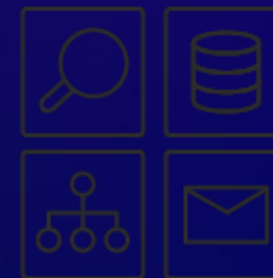
Top interests for customers new to AWS



Management and
governance

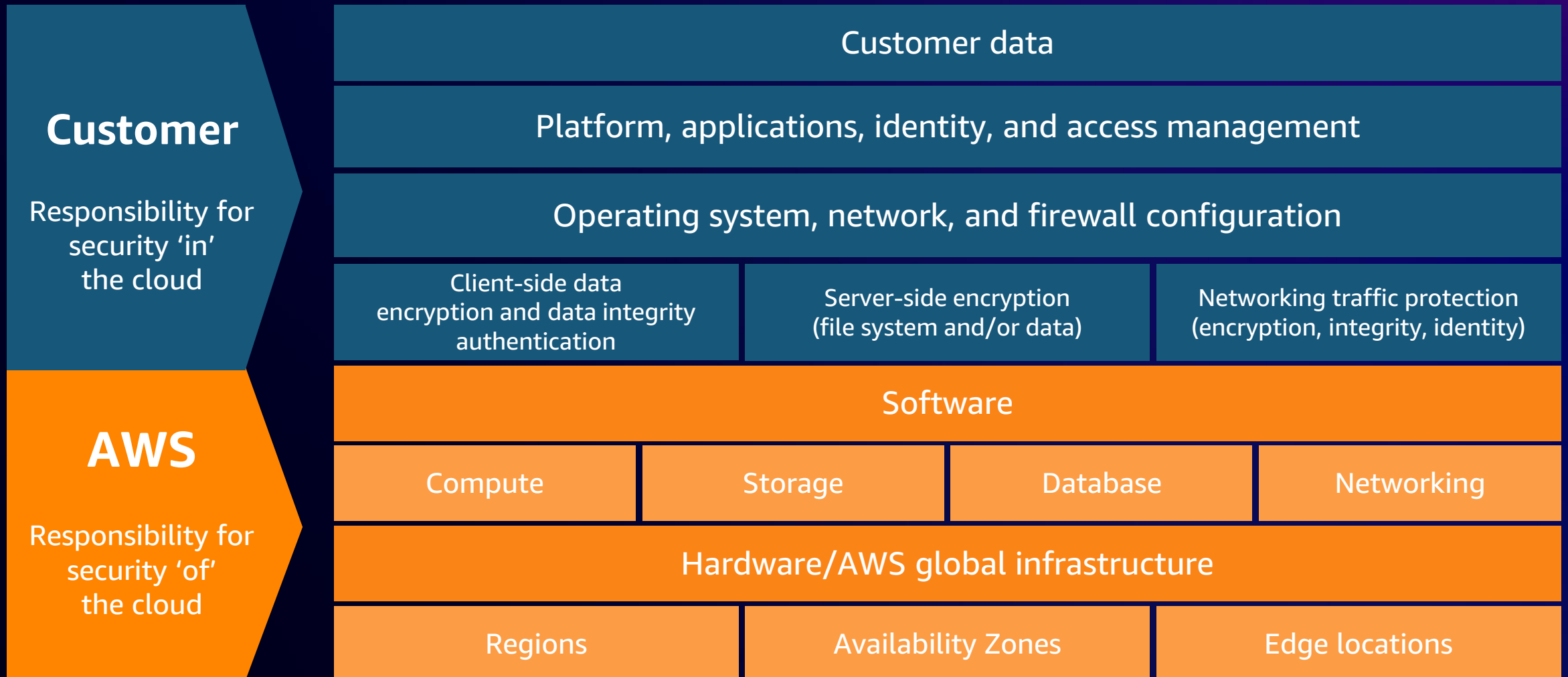


Security and
networking



Application
modernization

Shared responsibility model



Take advantage of security best practices

Follow the AWS Security Reference Architecture

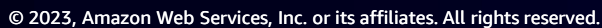
Includes

- Review of security foundations
- Diagrams, tooling options and descriptions for each level of the framework
- Review at this URL: a.co/eJQFHY1



Implement AWS Security Hub

- Pull together security findings from AWS and third-party sources
- Apply industry and government standards/controls
- Automate alerts and remediations



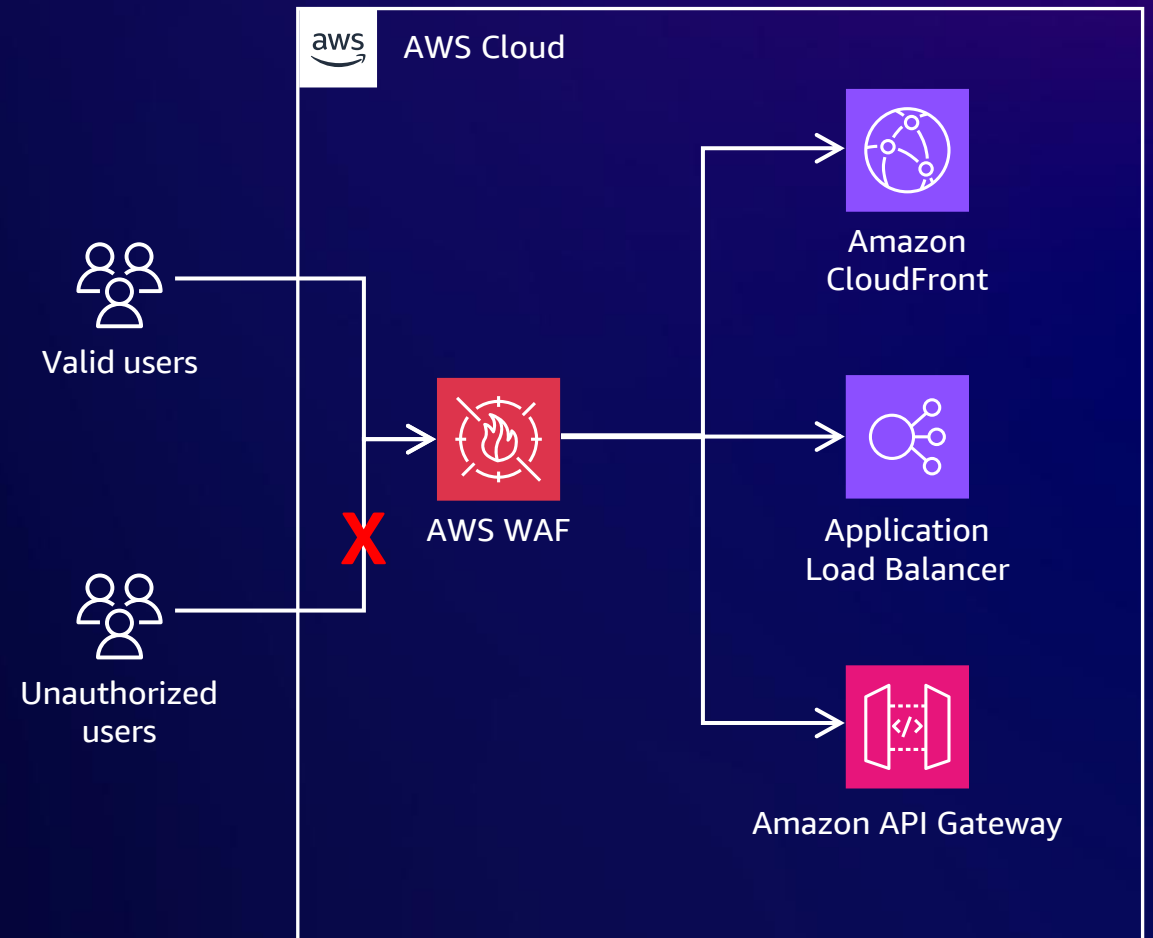
Secure all public-facing endpoints

Use AWS Web Application Firewall (WAF)

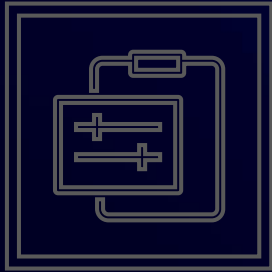
First line of defense against common exploits

Use built-in rules that cover (among other things) SQL Injection, XSS, and OWASP Top 10

Create your own rules, and use rules provided by third-party security vendors



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Application
modernization

Modernize with the strangler fig pattern

Gradually create a new system around the edges of the old monolith **until it is strangled**

- Keep the old monolith intact
- Create new microservices for new or replacement functionality
- Use API Gateway or Application Load Balancer to route traffic based on URL rules

AWS Migration Hub Refactor Spaces can make this easy

- Automatically provisions and manages networking components and routing rules, removing complexity for developers
- Learn more here: a.co/gmLSqm6



Use generative AI to build applications faster

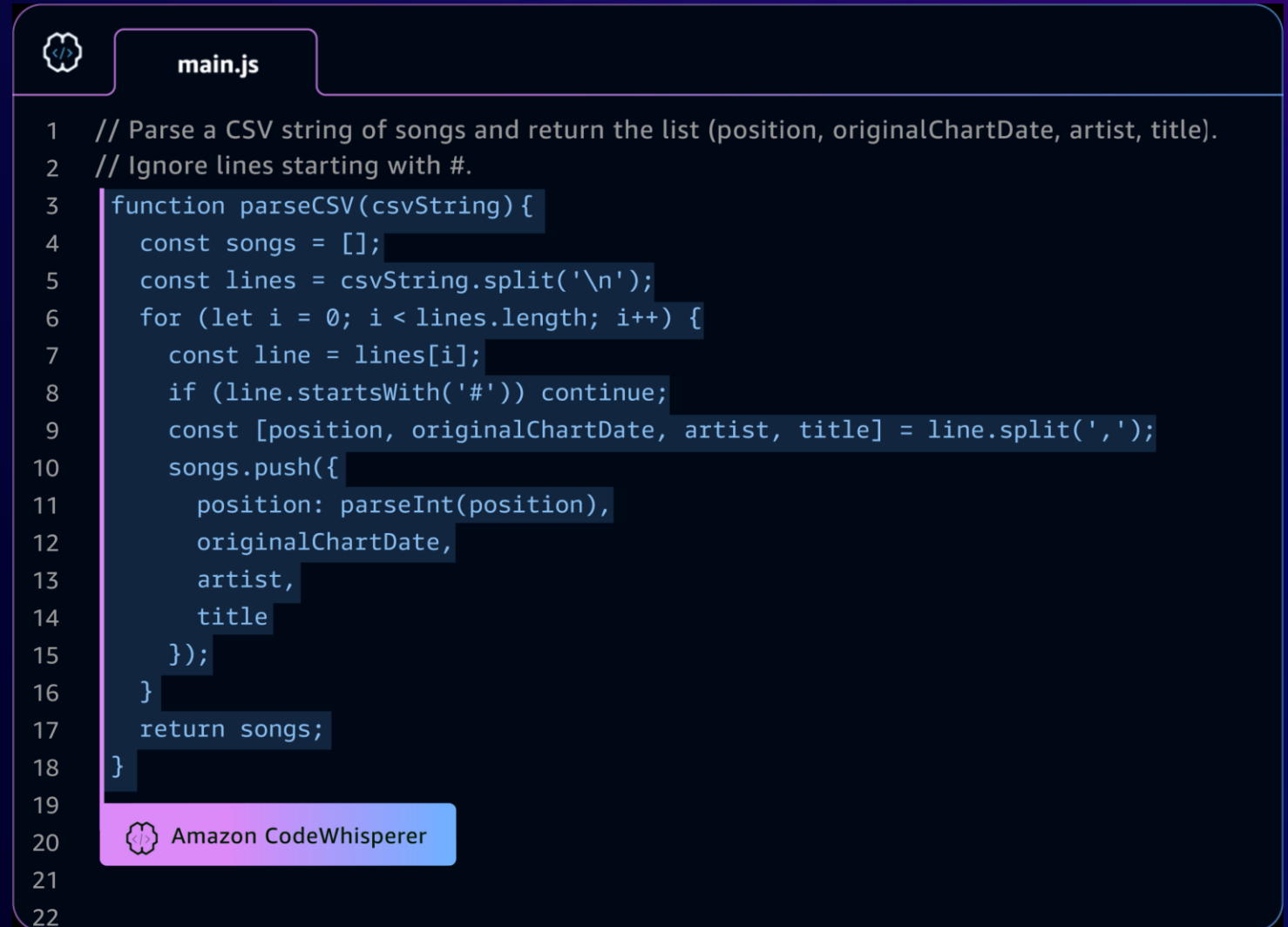
Amazon CodeWhisperer

Get multiple code suggestions in seconds based on natural language description of coding task and surrounding code

Generated code matches developer style and patterns

Scan generated and developer-written code to detect security vulnerabilities

Trained on billions of lines of code



The screenshot shows the Amazon CodeWhisperer interface. At the top, there's a tab labeled 'main.js'. Below it, a list of line numbers from 1 to 22 is visible. The code being edited is a JavaScript function 'parseCSV(csvString)' that takes a CSV string and returns an array of song objects. The function includes comments and logic to parse the CSV, skip lines starting with '#', and push parsed song data into an array. A blue suggestion box from Amazon CodeWhisperer is visible at the bottom, showing the function definition and its return statement.

```
1 // Parse a CSV string of songs and return the list (position, originalChartDate, artist, title).
2 // Ignore lines starting with #.
3 function parseCSV(csvString) {
4   const songs = [];
5   const lines = csvString.split('\n');
6   for (let i = 0; i < lines.length; i++) {
7     const line = lines[i];
8     if (line.startsWith('#')) continue;
9     const [position, originalChartDate, artist, title] = line.split(',');
10    songs.push({
11      position: parseInt(position),
12      originalChartDate,
13      artist,
14      title
15    });
16  }
17  return songs;
18 }
19
20 Amazon CodeWhisperer
21
22
```

Use cloud-based IDEs to avoid development environment headaches



AWS Cloud9

Cloud-based IDE that lets you write, run, and debug your code with just a browser

Simplify connections to private AWS resources, adding convenience in a sandbox when you are prototyping

Seamless experience for developing serverless applications, easily switching between local and remote code execution

Whiteboarding



Summary: Top 10 best practices for AWS beginners

Landing zone
(control tower)

Cost
management
(budgets)

Core networking

EC2 instance
management

Security
Reference
Architecture

Enable security
findings
(Security Hub)

Secure public
facing endpoints
(AWS WAF)

Modernization
strangler fig
pattern

Generative AI to
help build apps
(CodeWhisperer)

Cloud-based IDE
for developers
(Cloud9)

Thank you!

Keith Wood

kwoodaws@amazon.com

Renato Fichmann

fichmann@amazon.com



Please complete the session
survey in the mobile app

